

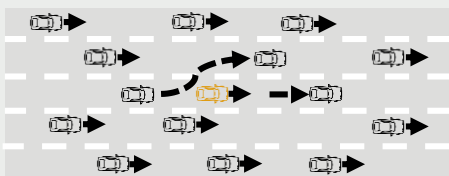
Condition Embedding

Text

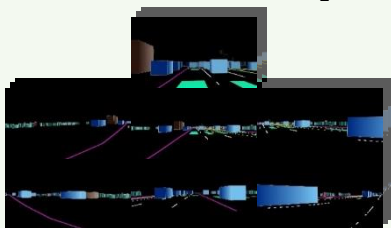
The scene unfolds under a bright, partly cloudy sky, with the sun casting a soft, diffused light across the urban landscape. The road is marked with clear white lines, including ...

Text
Encoder

Traffic Flow Generation Model



Multi-View HD Maps



Video
VAE
Encoder

Bidirectional DiT

Input Process Block

Patchify

Add HDmap Latent

DiT Block

DiT Block

DiT Block

DiT Block

DiT Block

...

DiT Block

Unpatchify

DiT Block

Self Attention

Adaptive Layer Norm

Cross Attention

Adaptive Layer Norm

FeedForward Network

Text Condition
Embedding

Denoised
Latent

Video
VAE
Decoder

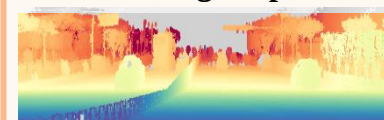
Multi-View RGB Videos



Denoised
Latent

LiDAR
VAE
Decoder

LiDAR Rangemap Videos



Denoising Variables

Multi-View RGB Images



Video
VAE
Encoder

Reference
Latent

Add Gaussian
Noise $\mathcal{N}(0, I)$

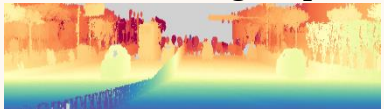
Noisy
Latent

Reference
Latent

Add Gaussian
Noise $\mathcal{N}(0, I)$

Noisy
Latent

LiDAR Rangemap



LiDAR
VAE
Encoder

LEGEND

Condition Process

Reference Data Process

Ego Vehicle

Surrounding Vehicle